Azetta.ai

Foundational AI Research Lab

Pioneering Interpretable and Efficient Intelligence through First-Principles Physics

THE PROBLEM: THE SCALING IMPASSE

Current deep learning methodologies prioritize **scaling size** above all else. This "brute force" approach leads to:

- **Bloated Architectures:** Models that are prohibitively expensive to train and deploy.
- **Non-Determinism:** Unpredictable outputs that lack a rigorous causal framework.
- **The "Black Box" Effect:** Emergent behaviors that are uninterpretable, necessitating costly "human-in-the-loop" intervention to reason through model behavior.

MATHEMATICAL FRONTIER: THE YAT-PRODUCT

We introduce a core shift in neural computation by replacing the standard dot-product interaction with the **First-Principled Gravitational Mercer Kernel**, also known as the **Taha Bouhsine Yat-Product** (ⵟ).

Standard Deep Learning Implementation

Traditional networks rely on:

$$y = \sigma(\mathbf{w} \cdot \mathbf{x} + b)$$

In this space, anti-parallel vectors are punished (-1), while orthogonal vectors are 0. Activation functions (like ReLU) must clip negative values to add non-linearity, leading to uninterpretable entanglement and information loss.

The Azetta Approach

Utilizing the Yat-kernel, we unify directional alignment and spatial proximity into a single, self-regulating operator:

$$K_{ⵟ}(\mathbf{w}, \mathbf{x}) = \frac{\langle \mathbf{w}, \mathbf{x} \rangle}{\|\mathbf{w} - \mathbf{x}\|^2 + \epsilon}$$

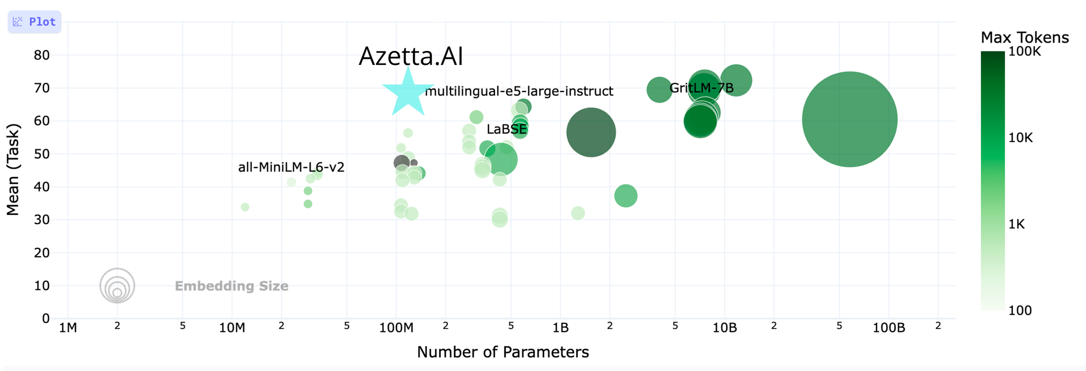
Our Angle: This provides **intrinsic non-linearity** without the need for clipping functions. The result is a fully interpretable latent space where models are:

- **10x Smaller** in total parameter count.
- **90% More Efficient** in indexing and inference.
- **White-Box by Design:** Every weight and gradient is inspectable and traces back to physical principles.

Performance Comparison

This demonstrates Azetta.AI's superior efficiency, achieving competitive performance (Mean Task Score ≈ 65) with only 100M parameters, compared to models requiring 500M+ parameters for similar results. This serves as a projected benchmark based on smaller models trained and benchmarked, as we have not yet officially benchmarked our scaled AI. We require funding for GPU compute infrastructure to train and validate a model of this scale:

Hugging Face MTEB Leaderboard: Performance per Model Size



GO-TO-MARKET PRODUCTS

Cosma DB: A high-performance Vector Database utilizing Yat-product based embeddings for near-instantaneous inference and lower memory overhead.

Omnilingual: Multilingual embedding models within a unified latent space, enabling direct cross-lingual semantic search without translation layers.

Periodica: A dedicated full-interpretability suite designed to monitor, explain, and audit Azetta AI deployments for safety-critical industries.

FUNDING ASK & REQUIREMENTS

We are raising a **$2M Pre-Seed Round** (Minimum check size: $50k) to fuel the next phase of foundational research and productization.

Allocation	Requirement
$1,000,000	5 World-Class Machine Learning Engineers
$700,000	GPU Compute and Infrastructure
$300,000	Legal, Operations, and GTM Strategy

TEAM

Our current team is led by **Taha Bouhsine**, **Douglas Seo**, and technical advisor **Jose Miguel Luna**.

COLLABORATION

Research conducted in collaboration with Professor **Krzysztof Choromanski** (Google DeepMind & Columbia University).

Contact: labs@azetta.ai / Confidential Executive Summary